

Node JS

Miscellaneous Tasks

[For Logical Reference]

Task-1

Write a node js script to write the text “You are creating a file” to help.txt file. After that append the text “you are appending data” to same help.txt file. After that read the file and print file contents on console. After finishing read operation, print the line “Thanks for using my program” on console. write, append,read sequence must be maintain. all read, write and append operations are asynchronous.

```
const fs = require('fs');
fs.writeFile('help.txt', 'You are creating a file', () => {
  fs.appendFile('help.txt', '\nyou are appending data', () => {
    fs.readFile('help.txt', 'utf-8', (err, data) => {
      console.log(data);
      console.log('Thanks for using my program');
    });
  });
});
```

Task - 2

Write a Node.Js program for following action

1. Write a file having five random elements separated by white space in .txt file.
2. append sorted array of these 5 elements in same file along with message : “Sorted array:” in new line.
3. Find maximum number from that and append with message “maximum number=” in same file.

```
const fs = require('fs');

fs.writeFileSync('num.txt', "20 30 13 7 22");
var d=fs.readFileSync("num.txt","utf-8");
d1= d.split(" ")

const sn = d1.sort((a, b) => a - b)
// Append sorted array message and sorted numbers
fs.appendFileSync('num.txt', `\nSorted array: ${sn}`);

const max = Math.max(...sn);
// Append maximum number message
fs.appendFileSync('num.txt', `\nmaximum number= ${max}`);
```

Task -3

Write a function 'ArrayToObject' which takes in an array of arrays, and returns an object with each pair of elements in the array as a key-value pair and store the result in one arraytoobject.txt file.

Input=[['Country', 'India'], ['State', 'Gujarat'], ['City', 'Ahmedabad']]

Output= { Country : ' India ', State : ' Gujarat ', City : 'Ahmedabad' }

```
function ArrayToObject(a) {
  var temp = {};
  temp[a[0][0]] = a[0][1]; // First key-value pair
  temp[a[1][0]] = a[1][1]; // Second key-value pair
  temp[a[2][0]] = a[2][1]; // Third key-value pair
  return temp;
}

var data = [['Country', 'India'], ['State', 'Gujarat'], ['City', 'Ahmedabad']];
var output = ArrayToObject(data);
console.log(output);

// Store the output in a file
const fs = require('fs');
fs.writeFileSync('arraytoobject.txt', JSON.stringify(output));
```

Task-4

write a node js Asynchronous program to perform CRUD operation of file management which should perform below task in sequence. **[Callback Hell]**

1. Create folder named "hello"
2. create a file in it named a.txt and add some data in it.
- 3.add more data at last in file
4. read data without getting buffer at first
5. rename the file
6. delete both the file and folder.

```
const fs = require('fs');
const additionalData = '\nMore data added at last.';

fs.mkdir("hello", (err) => {
  if (err) {
    console.error('Error creating folder:', err);
    return;
  }
  // Additional code for tasks 2-6 would go here
})
```

```
}
console.log('Folder created successfully.');
```

```
fs.writeFile("hello/a.txt", 'Some initial data.', (err) => {
  if (err) {
    console.error('Error creating file:', err);
    return;
  }
  console.log('File created successfully.');
```

```
fs.appendFile("hello/a.txt", additionalData, (err) => {
  if (err) {
    console.error('Error appending data:', err);
    return;
  }
  console.log('Data appended successfully.');
```

```
fs.readFile("hello/a.txt", 'utf8', (err, data) => {
  if (err) {
    console.error('Error reading file:', err);
    return;
  }
  console.log('File content:', data);
```

```
fs.rename("hello/a.txt", "hello/a1.txt", (err) => {
  if (err) {
    console.error('Error renaming file:', err);
    return;
  }
  console.log('File renamed successfully.');
```

```
fs.unlink("hello/a1.txt", (err) => {
  if (err) {
    console.error('Error deleting file:', err);
    return;
  }
  console.log('File deleted successfully.');
```

```
fs.rmdir("hello", (err) => {
  if (err) {
    console.error('Error deleting folder:', err);
    return;
  }
  console.log('Folder deleted successfully.');
```

```
});
});
});
});
});
});
});
```

Task-5

Create JSON object in file which contains array of objects. Calculate perimeter of square and perimeter of circle by using side value and diameter value respectively. And append final answers in file.

```
const shape = [ { name: "circle", diameter: 8}, {name: "square", side: 10}]

var ps=require("fs");

ps.writeFileSync("shape.txt",JSON.stringify(shape));
data=ps.readFileSync("shape.txt","utf-8");

b=JSON.parse(data);

if( b[0].name == 'circle'){
    var perimeter = (b[0].diameter/2) * 3.14 * 2 ;
    console.log(perimeter);
}
if ( b[1].name == 'square'){
    var peri = (b[1].side) *4 ;
    console.log(peri);
}
ps.appendFileSync("shape.txt","\nPerimeter of circle = "+ JSON.stringify(perimeter)+
"\nPerimeter of square = "+JSON.stringify(peri));
```

Output:

```
[{"name":"circle","diameter":8},{ "name":"square","side":10}]
Perimeter of circle = 25.12
Perimeter of square = 40
```

Task-6

Write node js script to perform tasks as asked.

- 1) Create one page with two links (Home(/) and about(/about)).**
- 2) Both pages must contain HTML type content and add required content on both the pages.**
- 3) If user add any other URL path, then he/she will be redirected to page and plain message will be displayed of "Page not found".**

```
var h=require("http");
var server=h.createServer(
    function(req,res)
    {
```

```

    if(req.url=="/")
    {
        res.writeHead(200,{"content-type":"text/html"});
        res.write("<h1> Home page </h1><div><ul><li><a href='/'>Home</a></li><li><a href='/about'>About</a></li></ul>");
        res.end();
    }
    else if(req.url=="/about")
    {
        res.writeHead(200,{"content-type":"text/html"});
        res.write("<h1> Home page </h1><div><ul><li><a href='/'>Home</a></li><li><a href='/about'>About</a></li></ul>");
        res.write("<h1> About Page </h1>");
        res.end();
    }
    else
    {
        res.writeHead(404,{"content-type":"text/plain"});
        res.write("Page not found");
        res.end("\nPlease check the url");
    }
  });
  server.listen(5051);
  console.log("Thanks!");

```

Task-7 (*Ref)

Create http webpage and display message “Welcome to Khushbu mam's class” in h1 tag after 10 seconds.

```

-----Client side Delay (Ref*) -----
const http = require('http');
// Create a server using the createServer method
const server = http.createServer((req, res) => {
  res.writeHead(200, {'Content-Type': 'text/html'});
  res.write(`
    <p id="demo"></p>
    <script>
      setTimeout(() => {
        let a = `<h2 style="color:tomato">Welcome to Khushbu mam's class</h2>`;
        document.getElementById("demo").innerHTML= <h2 style="color:tomato">Welcome to
        Khushbu mam's class</h2>
      },10000)

```

```

    </script>`)
    res.end();

});
const port = 3000;
server.listen(port, () => {
  console.log(`Server running at http://localhost:${port}/`);
});

```

-----Server Side Loading Delay-----

```

const http = require('http');

// Create a server using the createServer method

const server = http.createServer((req, res) => {

  setTimeout(() => {

    res.writeHead(200, {'Content-Type': 'text/html'});

    res.end('<h2 style="color:tomato"> Welcome to Khushbu mam\'s class</h2>');

  }, 10000); // 10000 milliseconds delay

});

const port = 3000;

server.listen(port, () => {

  console.log(`Server running at http://localhost:${port}/`);

});

```

Output Will Display after 10 second

Task-8

Create HTTP webpage on which home page will fetch json data, about page shows centrally aligned message “Hello from class” in red colour and cyan background using internal css styling. And any other page shows “404 error”.

(Render Response & Routing)

```

const http=require("http")
http.createServer((req,res)=>{

if(req.url==""){

```

```

const data={"name":"ABC","age":30}
res.writeHead(200,{"content-Type":"application/json"})
res.write(JSON.stringify(data))
res.end()
}
else if(req.url=="/about"){
res.writeHead(200,{"content-Type":"text/html"})

res.write(`<html>
      <head>
      <style>
      body { background-color: cyan;text-align: center;}
      h1 {color: red;}
      </style> </head>
    <body>
    <h1>Hello from class</h1>
    </body>
    </html>`)
res.end()
}
else{
res.writeHead(404,{"content-Type":"text/html"});
res.end("<h1>404 error</h1>")
}
}).listen(3000)
console.log("server running")

```

Task-9

Write node.js script to print “Welcome to Home Page” with two links containing two pages named as “About Us” and “Contact Us” on home page of server. If user request for About Us page it should display “Welcome to LJ University” in bold font-style with blue color and if user request for Contact Us page it should display “Email:abc@ljinstitutes.edu.in” in italic font-style with red color if any other request is requested it shows “Page not found” message in plaintext.

```

const http = require('http');
const server = http.createServer((req, res) => {
  res.writeHead(200, { 'Content-Type': 'text/html' });

  if (req.url === '/') {
    res.end(`
      <h1>Welcome to Home Page</h1>
      <a href="/about">About Us</a>  <a href="/contact">Contact Us</a>
    `);
  } else if (req.url === "/about") {
    res.end(`<h1 style="color: blue; font-weight: bold;">Welcome to LJ University</h1>`);
  }
});

```

```
    } else if (req.url === "/contact") {
        res.end(`<h2 style="color: red; font-style: italic;">Email: abc@ljinstitutes.edu.in</h2>`);
    } else {
        res.writeHead(404, { 'Content-Type': 'text/plain' });
        res.end("Page not found");
    }
});
server.listen(3000, () => {console.log("Server started");});
```

Task-10

Create HTTP webpage on which Home page display “Welcome to Log in page” in blue color using inline CSS, Login page shows one Form with detail for Username, Password, submit button On any other page “Page not found” message has to appear.

```
const http = require("http");
const fs = require("fs");
const server = http.createServer((req, res) => {
    if (req.url === "/") {
        res.writeHead(200, { "Content-Type": "text/html" });
        res.write(`<h1 style="color:blue">Welcome to Log in page</h1>`);
        res.end();
    }
    else if (req.url === "/login") {
        res.writeHead(200, { "Content-Type": "text/html" });
        res.write(`<form action="#" method="POST">
            Username: <input type="text" name="username"><br><br>
            Password: <input type="password" name="password"><br><br>
            <input type="submit" value="Submit">
        </form>`);
        res.end();
    }
    else {
        res.writeHead(404, { "Content-Type": "text/html" });
        res.write("<h1>Page Not Found</h1>");
        res.end();
    }
});
server.listen(5001)
```


Task-11

Create HTTP webpage on which Home page display “Welcome to Log in page” in blue color and font size must be 32px, Login page shows one HTML file from static URL having Form with detail for Username, Password, submit and reset button, Gallery page reflect one Image “hello.jpg” and any other page shows “Page Not found”. Write all necessary files to perform task. (Image already exist in same folder)

login.js

```
var h = require("http");
var url = require("url");
var fs = require("fs");
var addr="http://localhost:5051/login.html";
var server = h.createServer(function (req, res) {
  if (req.url == "/" ) {
    res.writeHead(200, { "Content-Type": "text/html" });
    res.write("<h1 style='color:blue;font-size:32px;'>Welcome to login page</h1>");
    res.end();
  } else if (req.url == "/login") {
    var q=url.parse(addr,true);
    data=fs.readFileSync("."+q.pathname);
    res.writeHead(200, { "Content-Type": "text/html" });
    res.write(data);
    res.end();
  } else if (req.url == "/gallery") {
    img = fs.readFileSync("1.png");
    res.writeHead(200, { "Content-Type": "image/png" });
    res.end(img); // Sending image data
  } else {
    res.writeHead(404, { "Content-Type": "text/plain" });
    res.write("Page not found");
    res.end("\nPlease check the URL");
  }
});
server.listen(5051)
```

login.html

```
<html>
<body>
  <h2>Login Form</h2>
  <form action="#" method="POST">
    <label for="username">Username:</label>
    <input type="text" required><br><br>
```

```
<label for="password">Password:</label>
<input type="password" required><br><br>

<button type="submit">Submit</button>
<button type="reset">Reset</button>
</form>
</body>
</html>
```

Task-12

Write node js script to fetch values from url given below and display output as asked.
"https://www.google.com/exam.txt?c1=Hello&c2=FSD2 T1 Test&c3=Welcome to LJU#AllTheBest"

1) Data must be written as below in file named "exam.txt". File name must be fetched from the url given above.

Output:

Hello!

Welcome to LJU FSD2 T1 Test

#AllTheBest

2) Read content from file "exam.txt" and send response to server and display data in "/" page in same format as above but in H1 tag and in red color.

3) If any other page is requested it shows "Page not found" message in plain text.

```
const http = require("http");
const fs = require("fs");
const url = require("url");

const url1 =
"https://www.google.com/exam1.txt?c1=Hello&c2=FSD2+T1+Test&c3=Welcome+to+LJU#A
llTheBest";
const parsedUrl = url.parse(url1,true);
console.log(parsedUrl.pathname)

const c1 = parsedUrl.query.c1;
const c2 = parsedUrl.query.c2;
const c3 = parsedUrl.query.c3;
const hash = parsedUrl.hash;
const filename = "."+parsedUrl.pathname

const fileContent = c1+"!\n"+c3+" "+c2+"\n"+hash;
```

```
fs.writeFileSync(filename, fileContent);
const server = http.createServer((req, res) => {
  if (req.url === '/') {
    data= fs.readFileSync(filename, "utf-8")
    res.writeHead(500, { "Content-Type": "text/html" });
    res.end("<h1 style='color:red'><pre>"+data+"<pre></h1>");
  }
  else {
    res.writeHead(404, { "Content-Type": "text/plain" });
    res.end("Page not found");
  }
});
server.listen(3000)
```

Task-13

Write a nodeJS script to fire an event named calculate which calculates the total marks of 5 subjects about of 25 marks and displays the total marks on console as an output. The calculate event fires another event name percentage which takes total marks as argument and percentage should get displayed in console.

```
const EventEmitter = require("events");
const ee = new EventEmitter();

// Event to calculate total marks
ee.on("calculate", () => {
  const marks = [22, 18, 20, 25, 24];
  let total = 0;
  for (let i = 0; i < marks.length; i++) {
    total += marks[i];
  }
  console.log("Total Marks:", total);
  ee.emit("percentage", total);
});

ee.on("percentage", (total) => {
  const totalMaxMarks = 5 * 25;
  const percentage = (total / totalMaxMarks) * 100;
  console.log("Percentage:", percentage + "%");
});
ee.emit("calculate");
```

Task-14

Write a NodeJS code using the Event module to simulate a user registration system. The program should trigger an event by passing a username and a value indicating whether the user is “fresh” or “experienced”. When the event is handled, it should display a welcome message including the username. (for example, if the username is abc, the message should be Welcome abc). If the user is “fresh”, the program should trigger another event that generates and displays an email by concatenating the username with @insti.co.in (for example, if the username is abc, the email should be abc@insti.co.in). If the user is “experienced”, only the welcome message should be displayed. Ensure that the events are properly handled and executed in the correct order.

```
const EventEmitter = require("events");
const event = new EventEmitter();
event.on("registerUser", (username, status) => {
  console.log(`Welcome ${username}`);
  if (status === "fresh") {
    event.emit("generateEmail", username);
  }
});
event.on("generateEmail", (username) => {
  const email = username + "@insti.co.in";
  console.log(`Generated Email: ${email}`);
});
event.emit("registerUser", "abc", "fresh");
```

Task-15

Write a NodeJS code to create a folder named tempReports and inside it, create a file called daily.txt containing the initial text "Report Start". Then, it should add three entries to the file: "Entry 1: Success", "Entry 2: Warning", and "Entry 3: Success". After writing, the program must read the file and count how many times the word "Success" appears. If the word "Success" appears three or more times, the program should delete the tempReports folder. Ensure all file operations are asynchronous.

```
const fs = require('fs');

const folderName = 'tempReports';
```

Write a NodeJS code to calculate and display the profit or loss incurred by an investor

```
const http = require('http');
const quantity = 100;
const buyPrice = 50;
const sellPrice = 60;
const investment = quantity * buyPrice;
const returnAmount = quantity * sellPrice;
const result = returnAmount - investment;
const server = http.createServer((req, res) => {
  res.writeHead(200, { 'Content-Type': 'text/html' });

  let message = "";
  let color = "";
  if (result > 0) {
    message = `You made a profit of Rs.${result}`;
    color = 'green';
  } else if (result < 0) {
    message = `You incurred a loss of Rs.${Math.abs(result)}`;
    color = 'red';
  } else {
    message = 'No profit, no loss.';
    color = 'black';
  }
  res.write(`<h1 style="color: ${color};">${message}</h1>`);
  res.end();
});
server.listen(3000, () => {
  console.log('Server is running at http://localhost:3000');
});
```

Task-17

Write a Node.js program to create a web server that handles two pages. When a user visits the Home page, the server should respond with "Welcome to Home Page", which must be styled using an external CSS file so that the text appears in blue color and is center aligned. On any other page, it should display page not found.

Style.css

```
h1 {
  color: blue;
  text-align: center;
}
```

Task.js

```
const http = require("http");
const fs = require("fs");

http.createServer((req, res) => {
  if (req.url === "/" ) {
    res.writeHead(200, { "Content-Type": "text/html" });
```

```
res.end(`
  <head>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <h1>Welcome to Home Page</h1>
  </body>
`);
}
else if (req.url === "/style.css") {
  data=fs.readFileSync("style.css")
  res.end(data)
}
else {
  // 404 Page
  res.writeHead(404, { "Content-Type": "text/html" });
  res.end("<h1 style='color:red;'>Page Not Found</h1>");
}
}).listen(3000);
```

Task-18

Write a script to define two JSON objects named as “division1” and “division2” having an array to store 5 integer numbers. Write this object in a file named XYZ.txt using file system. Define third array as result which gives output as explained below: Suppose first array contains base value and second array contains power value and then third array will give output as: a^b . For example: first array[0]=2 and second array[0]=3 then it should return 8 in third array[0]. Also append this result in XYZ.txt and as well as on console.

```
const fs = require('fs');

// Define two JSON objects
const division1 = {
  numbers: [2, 3, 4, 5, 6] // Base values
};

const division2 = {
  numbers: [3, 2, 1, 2, 3] // Power values
};

// Write initial objects to file
const initialData = {
  division1: division1,
  division2: division2
};

fs.writeFileSync('XYZ.txt', JSON.stringify(initialData, null, 2));

// Calculate result array ( $a^b$ )
```

```
let result = [];  
  
for (let i = 0; i < division1.numbers.length; i++) {  
    let base = division1.numbers[i];  
    let power = division2.numbers[i];  
    result.push(Math.pow(base, power));  
}  
  
// Append result to file  
const resultData = "\n\nResult Array:\n" + JSON.stringify(result);  
  
fs.appendFileSync('XYZ.txt', resultData);  
  
// Display result on console  
console.log("Result Array:", result);
```

Task-19

Write a Node.js program using the HTTP module to create a server that sends a JSON response when the user visits the /api route. The JSON object should contain: name as "Rahul", course as "NodeJS", status as "Enrolled" Set the appropriate header for JSON data and display "404 – Page Not Found" for any other route.

```
const http = require('http');  
  
// Create server  
const server = http.createServer((req, res) => {  
  
    // Check route  
    if (req.url === '/api') {  
  
        // Set header for JSON response  
        res.writeHead(200, { 'Content-Type': 'application/json' });  
  
        // Create JSON object  
        const data = {  
            name: "Rahul",  
            course: "NodeJS",  
            status: "Enrolled"  
        };  
  
        // Send JSON response  
        res.end(JSON.stringify(data));  
    }  
    else {  
        // Handle other routes (404)  
        res.writeHead(404, { 'Content-Type': 'text/plain' });  
        res.end("404 - Page Not Found");  
    }  
});
```



```
// Run server
server.listen(3000, () => {
  console.log("Server running at http://localhost:3000");
});
```

Task-20

Write code to perform the tasks as asked below.

1. Add three buttons.
2. Increase button to increase the fonts. It should stop increasing the fonts when the font size reaches to 200px or stop button is clicked.
3. Stop button to stop increasing or decreasing the fonts.
4. Decrease button to decrease the fonts. It should stop decreasing the fonts when the font size reaches to 20px or stop button is clicked.
5. Increasing/decreasing interval is of 100 ms and Default font size = 50px

```
<body>
  <p id="p1" style="font-size: 50px;">Hello</p>
  <button onclick="changeFont(1)">Increase</button>
  <button onclick="changeFont(-1)">Decrease</button>
  <button onclick="stopped = true">Stop</button>

  <script>
    let font = 50, stopped = false, direction = 0;

    function changeFont(d) {
      if (stopped) { direction = d; stopped = false; return; }
      if ((d > 0 && font < 100) || (d < 0 && font > 15)) {
        font += d;
        document.getElementById('p1').style.fontSize = font + 'px';
        setTimeout(() => changeFont(d), 100);
      }
    }
  </script>
</body>
```